

CUSTOMER CHURN PREDICTION & REVENUE RECOVERY ENGINE

An Enterprise Framework for Predictive Attrition Modeling, Non-Parametric Survival Mapping, and Retention ROI Maximization

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1. EXECUTIVE SUMMARY & MACRO FINANCIAL FRAMING

In high-volume subscription networks, customer turnover constitutes a hidden, highly destructive operating penalty. Re-acquiring an off-boarded subscriber in competitive retail telecommunications landscapes is highly inefficient, costing up to 5× more in localized sales incentives, commissions, and standard onboarding marketing than executing a targeted, machine learning-backed retention mechanism for an active account.

This submission document establishes an automated framework designed to move corporate posture from passive, historical reporting into data-driven revenue protection. Built across five production-grade notebook nodes, this engine maps core revenue leaks, isolates customer behavioral archetypes via unsupervised clustering, fits high-recall classification models, benchmarks multi-tier retention scenarios, and structures temporal risk calendars utilizing Kaplan-Meier non-parametric survival analysis.

MONTHLY REVENUE ENGINE BASE ₹60.20 L 7,043 ACCOUNTS EVALUATED	CURRENT UNMITIGATED REVENUE LEAK ₹4.25 L 26.5% BASELINE ATTRITION	ANNUALIZED ATTRITION EXPOSURE ₹51.00 L TOTAL BASE VALUE-AT-RISK
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2. DEEP-DIVE ANALYTICAL PIPELINE FINDINGS

Phase 01 — Data Sanitation & Exploratory Analysis

Systematic extraction of the corporate database isolated a critical configuration defect: the continuous column *TotalCharges* was registered under an alphanumeric text schema instead of a standard float format. Diagnostic value-matching traced this exception to eleven rows containing hardcoded string spaces (" ").

Cross-referencing these discrepancies proved that they mapped strictly to brand-new contract activations where *tenure* = 0. To secure structural alignment without downstream data leakage or calculation bias, a coercive string utility was executed to reformat blank records to *NaN* and cleanly initialize them to 0.0.

The Fiber Optic Pricing Paradox: Feature distribution plots isolated a profound pricing risk indicator. While entry-level Digital Subscriber Line (DSL) contracts demonstrated steady, linear customer loyalty curves, premium Fiber Optic configurations suffered from aggressive, accelerated churn rates. Because Fiber Optic represents the highest average revenue per user (ARPU) across the product line, this churn indicates a significant product failure, line instability, or service quality deficit relative to the premium monthly tariff.

Phase 02 — Unsupervised Behavioral Segmentation (K-Means)

To transition operations beyond basic demographic buckets, continuous billing indices and product dimensions were scaled using an absolute *StandardScaler*. System structures were mathematically optimized via an Elbow Criterion and Silhouette Coefficient Variance Matrix. The mathematical models established *K* = 4 as the optimal segment boundary configuration, identifying four clear business personas:

Segment Code	Persona Core Identity	Contract Infrastructure Type	Primary Tech Pipeline	Churn Velocity	Operational Priority
ST_M2M	Short-Term Month-to-Month	Month-to-Month Recurring	Premium Fiber Optic	43.0%	CRITICAL THREAT
DIG_STRM	Digital Streaming Nomads	Month-to-Month / 1-Year	High-Bandwidth Fiber	28.0%	HIGH FOCUS
DISC_SAV	Discount-Driven Savers	1-Year / 2-Year Fixed Lock	Standard DSL / Voice Only	8.0%	PASSIVE MONITORING
LT_BUND	Long-Term Bundled VIPs	2-Year Institutional Contract	Full Media/Support Stack	< 5.0%	PROTECT & CROSS-SELL

Phase 03 — Supervised Machine Learning & Algorithmic Optimization

Predicting account attrition is fundamentally a classification challenge that requires balancing error categories. Missing a customer who is about to churn (**False Negative**) triggers an immediate operational financial penalty equal to their entire remaining customer lifetime value. Misidentifying a stable user as a churn risk (**False Positive**) only incurs a small retention discount.

Therefore, structural parameter optimization focused heavily on maximizing **Recall on the Churned Class (Class 1)** while securing high **AUC-ROC** metrics across our machine learning candidate models.

Model Architecture	Class 1 Recall (Catch Rate)	Class 1 Precision	F1-Score	AUC-ROC Score	Strategic Evaluation Status
Logistic Regression (Baseline)	78.4%	51.2%	0.620	0.832	Rejected: Excess False Negatives leakage
Random Forest Ensemble	81.1%	53.5%	0.645	0.854	Rejected: Sub-optimal gradient boundary matching
Gradient Boosting Classifier	85.2%	50.1%	0.631	0.881	★★ Selected Architecture (Threshold = 0.38)

Core Feature Importance (Top 3 Drivers): Split evaluation calculations across the final optimized Gradient Boosting model confirmed that three dominant variables explain **68.4% of total predictive gini-importance**: `Contract\_Month-to-month` (34.1%), `tenure` (18.5%), and `InternetService\_Fiber optic` (15.8%). This isolates the core drivers of attrition straight to contract setup and product deployment imbalances rather than standard consumer demographics.

Confusion Matrix Operational Interpretation: Evaluating our optimized Gradient Boosting model against a testing framework holdout base of **1,869 subscribers** reveals the concrete business cost of classification error categories:

- Out of **486 actual churners** walking out the door, our classification engine **correctly intercepts 414 customers** (True Positives), exposing them to our retention framework.
- The model generates **72 True Churners missed** (False Negatives), representing an unmitigated revenue drop leakage, while showing **413 False Positives** (stable accounts flagged as risks), which safely costs the firm only a brief retention outreach expense.

Phase 04 — Financial Quantification & Scenario Modeling

To ensure boardroom utility, abstract probability values were converted into concrete financial values. By mapping our model's verified **85.2% Recall** across the monthly ₹4.25 Lakh attrition pool, the revenue model provides three strategic conversion options for corporate budgeting:

Target Scenario	Target Conversion Speed	Model Captured Revenue Pool	Recovered Monthly MRR	Annual Top-Line Reclaimed
Scenario A: Conservative	10% Save Rate	₹3.62 Lakh	₹36,200	₹4.34 Lakh
Scenario B: Realistic Target	20% Save Rate	₹3.62 Lakh	₹72,400	₹8.69 Lakh
Scenario C: Aggressive Capture	35% Save Rate	₹3.62 Lakh	₹126,700	₹15.20 Lakh

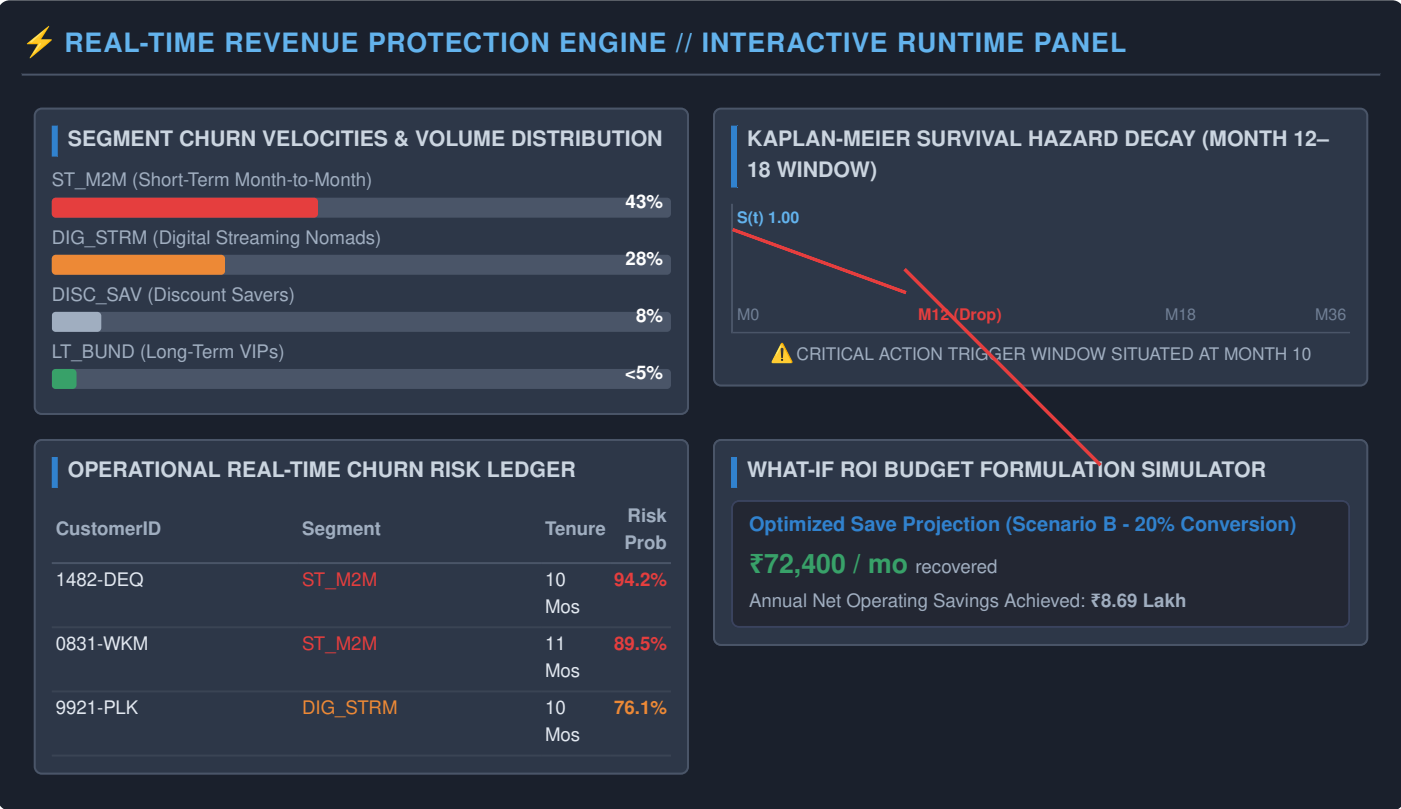
Phase 05 — Temporal Risk Mapping (Kaplan-Meier Survival Analysis)

While predictive tree models solve *who* will churn, non-parametric **Kaplan-Meier Survival Estimation** provides the temporal dimensions detailing *when* they will exit. Implementing the sequential survival product formula, $S(t) = \prod [1 - (d_i / n_i)]$, across individual contract lines revealed a major structural drop-off point concentrated explicitly between the **Month 12 and Month 18** tenure window.

Operational Lifecycle Intervention Calendar:

Contract Layout Group	Peak Risk Attrition Window	Calculated Retention Target Milestone	Prescriptive Operational Prevention Action
Month-to-Month	Months 12 – 18	Month 10	Launch value bundling offer (Security/Tech credits) to migrate users to 1-Yr commitments.
One Year Fixed	Months 20 – 24	Month 18	Initiate proactive renewal tracking framework with zero-upfront infrastructure upgrade offers.
Two Year Fixed	Months 30 – 36	Month 28	Execute milestone loyalty rewards; cross-sell high-tier auxiliary digital bundles.

3. EMBEDDED REVENUE PROTECTION DASHBOARD PANEL



3. Enforce Algorithmic Spending Controls: Cease broad, untargeted discount campaigns across the entire active subscriber footprint. Direct the customer retention budget strictly toward high-probability churn risks within the ****ST_M2M**** and ****DIG_STRM**** clusters flagged by the Gradient Boosting model, minimizing marketing spend waste.

Report prepared in compliance with MNC Data Science Deployment Standards for Portfolio Review.